

When does the PAN enter the database?

At upload. Both PAN columns are written together, in one database write, while the “Parsing...” spinner is on screen. Confirm Import only makes that PAN permanent.

- The answer
- Just Me flow
- Family flow
- Upload → Confirm timeline
- PAN column lifecycle
- ER diagram
- Where PAN is stored

The three PAN-related columns, screen by screen

All three live on `household_members` . `pan_pending_until` holds no PAN: it only says whether the PAN is still pending.

1 · MEMBER CREATED		
Just Me screen / Add Family Members		
pan_encrypted		NULL
pan_lookup_hash		NULL
pan_pending_until		NULL
A row exists for the person, with no PAN yet.		
2 · UPLOAD ACCEPTED		
“Parsing...” spinner		
pan_encrypted		encrypted PAN
pan_lookup_hash		HMAC hash
pan_pending_until		now + 65 min
The PAN enters the database here. One UPDATE and one COMMIT, during <code>POST /imports/parse</code> .		
3 · REVIEWING		
Review CAS Import screen		
pan_encrypted		encrypted PAN
pan_lookup_hash		HMAC hash
pan_pending_until		counting down

Leaving here (Try again / Cancel, or 65 min pass) wipes all three.

4 · CONFIRM IMPORT CLICKED

“Import complete”

pan_encrypted	unchanged
pan_lookup_hash	unchanged
pan_pending_until	NULL = permanent

Confirm only clears the pending flag. It also saves the import, folios, transactions and the CAS PDF.

Never one without the other

pan_encrypted and pan_lookup_hash are always saved in the same write.

Before upload: memory only

Until then the raw PAN exists only in server memory, never on disk in plain form.

Conflict = nothing written

If the PAN belongs to someone else, a popup appears and the database is untouched.

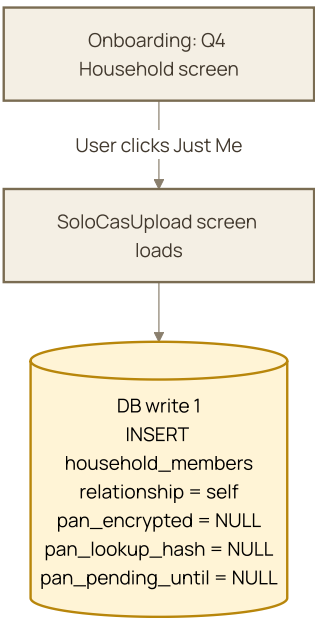
Family: one member at a time

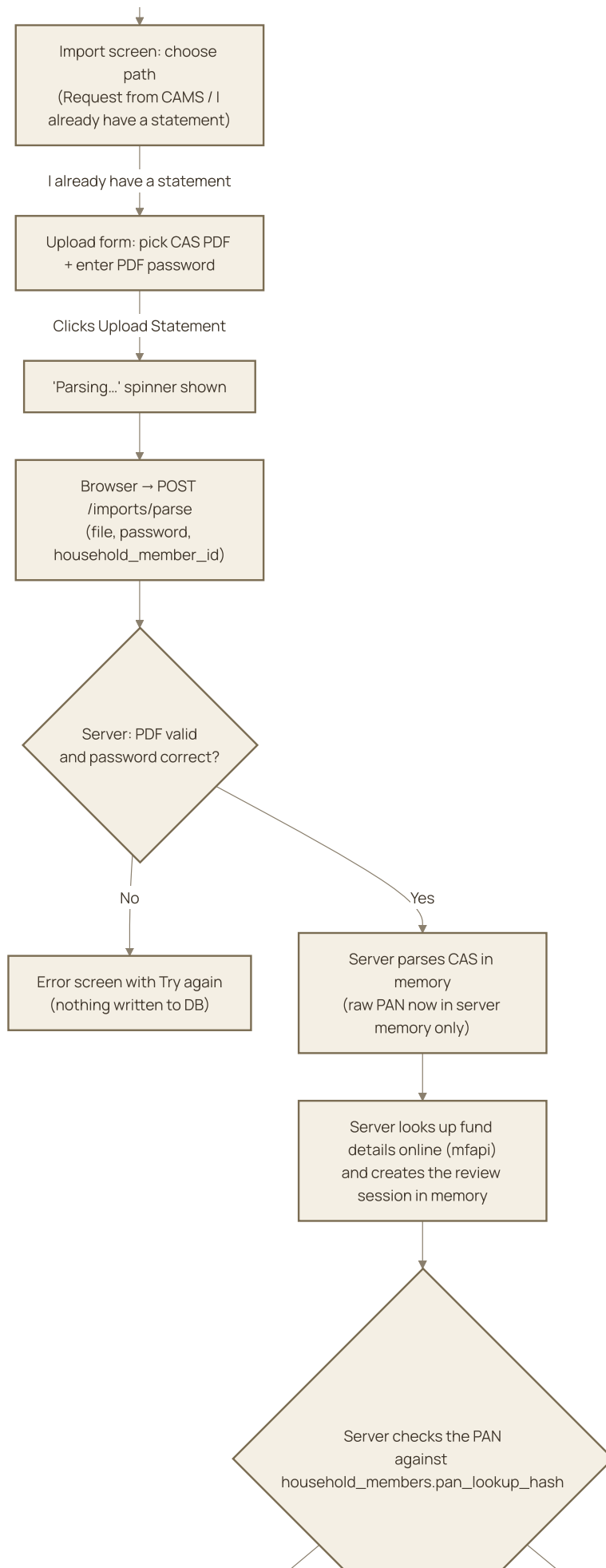
Picking files only queues them in the browser. Each PAN is written after “Import now”, in turn.

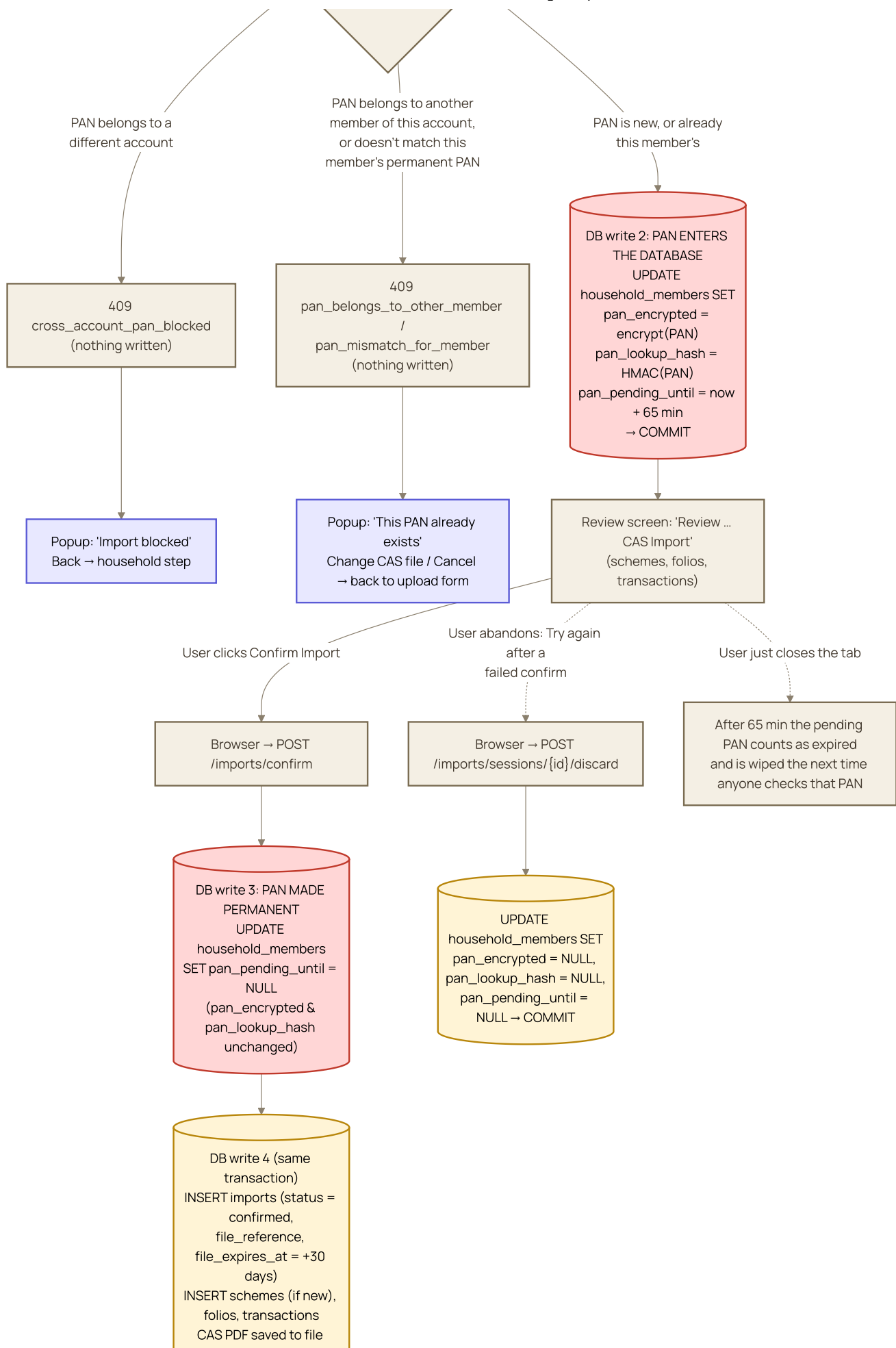
Just Me: from clicking “Just Me” to “Import complete”

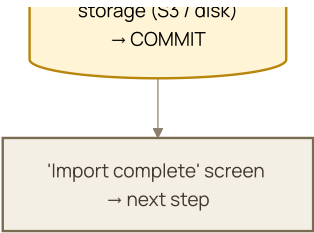
Every screen, server step and database write, in order. The red boxes are the two moments the PAN columns change.

☐ PAN columns change ☐ Other database writes ☐ Popup shown







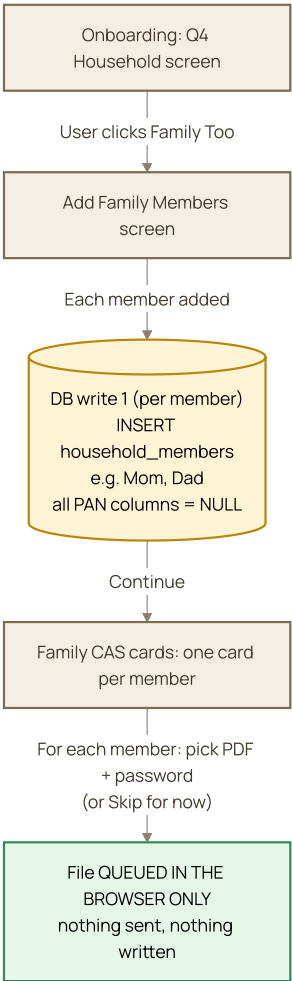


In a fresh Just Me account the only popup that can realistically appear is **Import blocked**, when the same CAS was already imported under a different Unifolio account.

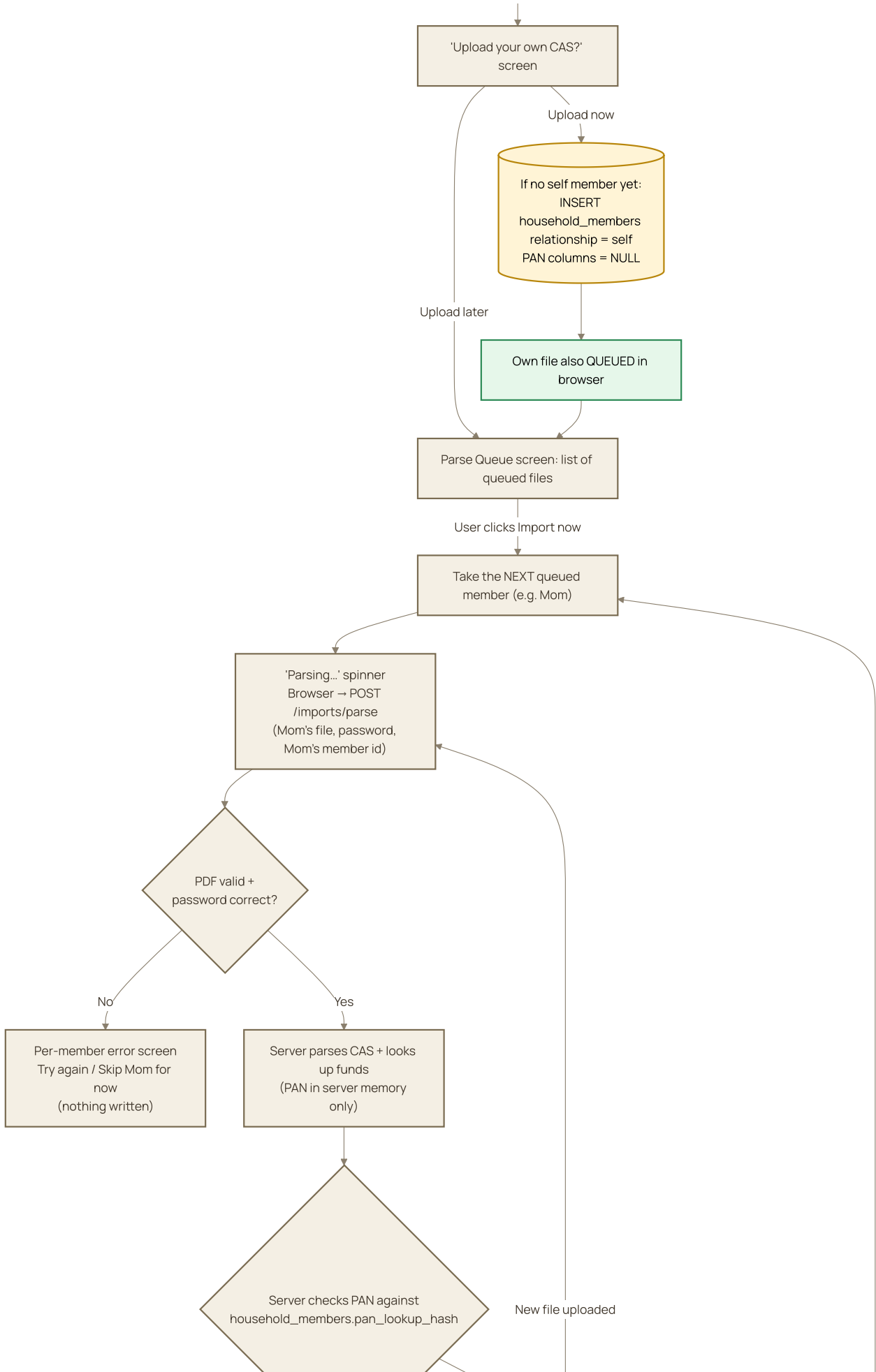
Family: files are queued, then each member is uploaded in turn

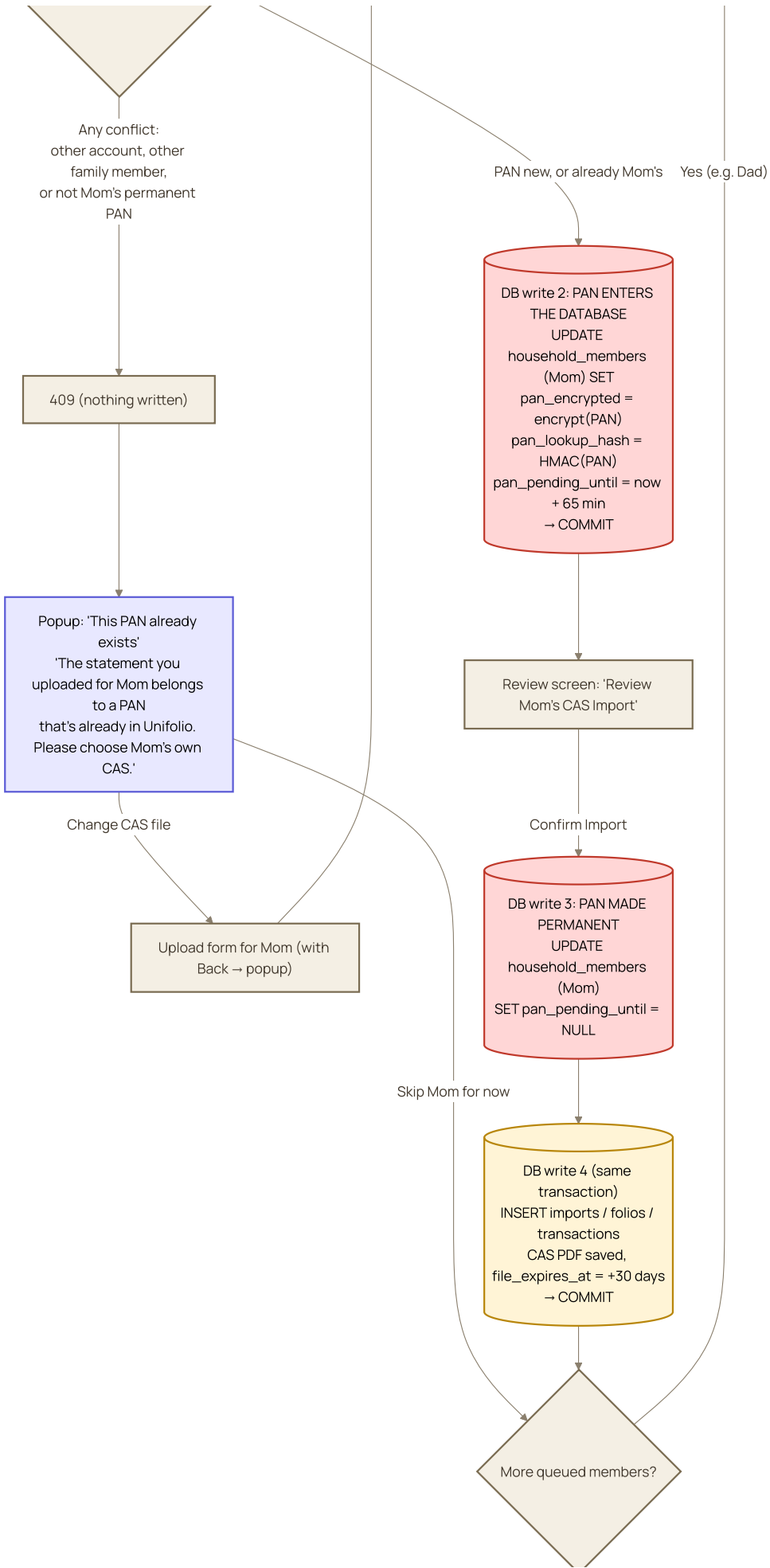
Picking files on the member cards sends nothing to the server. Each member's PAN is checked and written only after "Import now".

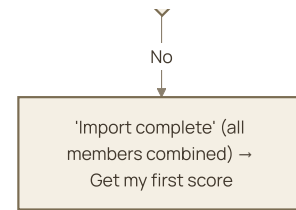
- PAN columns change
- Other database writes
- Popup shown
- Browser only, nothing sent



PAN Storage Map



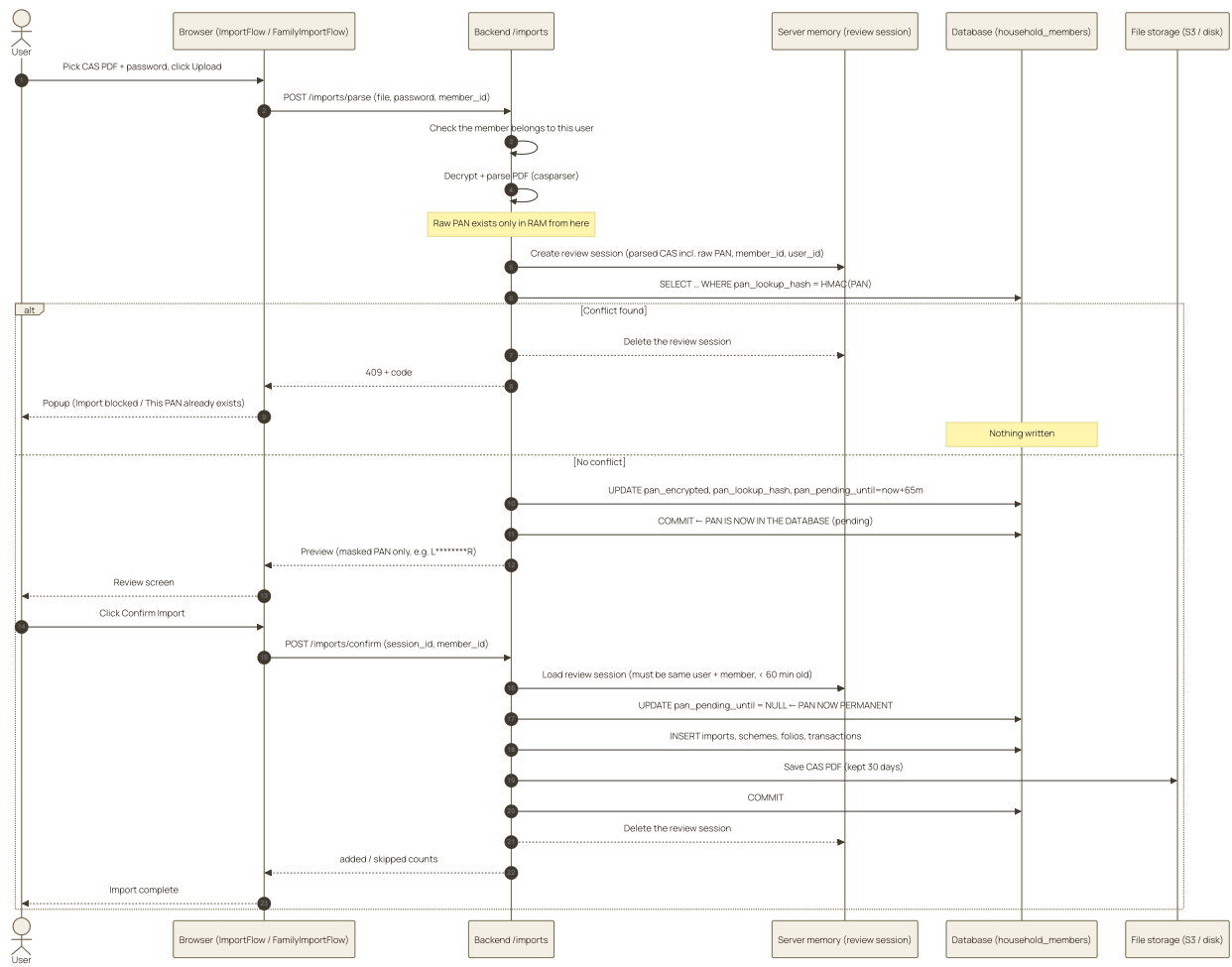




Why the order matters: Mom's PAN is permanent before Dad's file is even uploaded. If Dad's slot holds Mom's statement by mistake, Dad's upload hits the "This PAN already exists" popup.

One upload → confirm, step by step

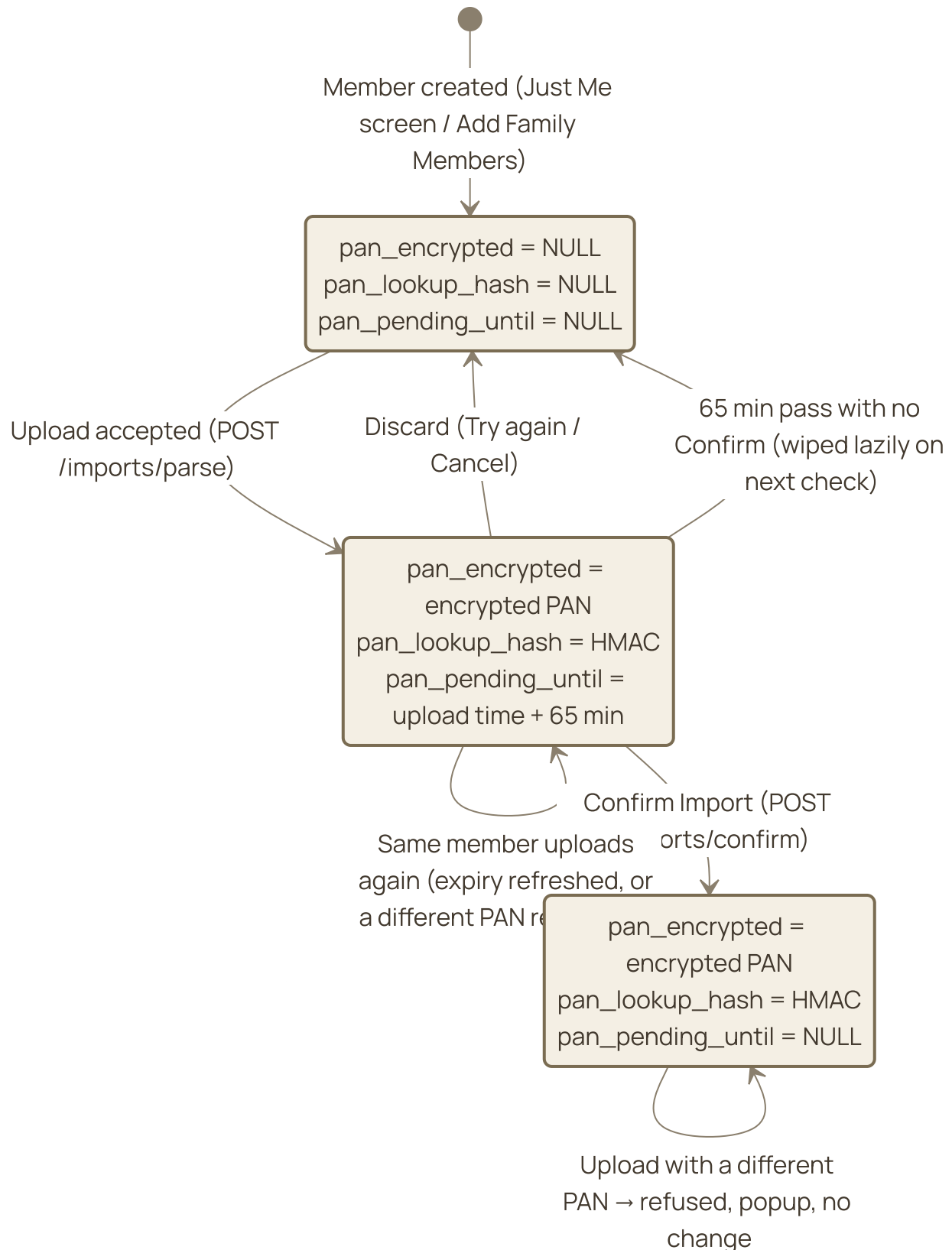
What sits in server memory versus the database at each step.



Lifecycle of the PAN columns on one member

Lifecycle of the PAN columns on one member

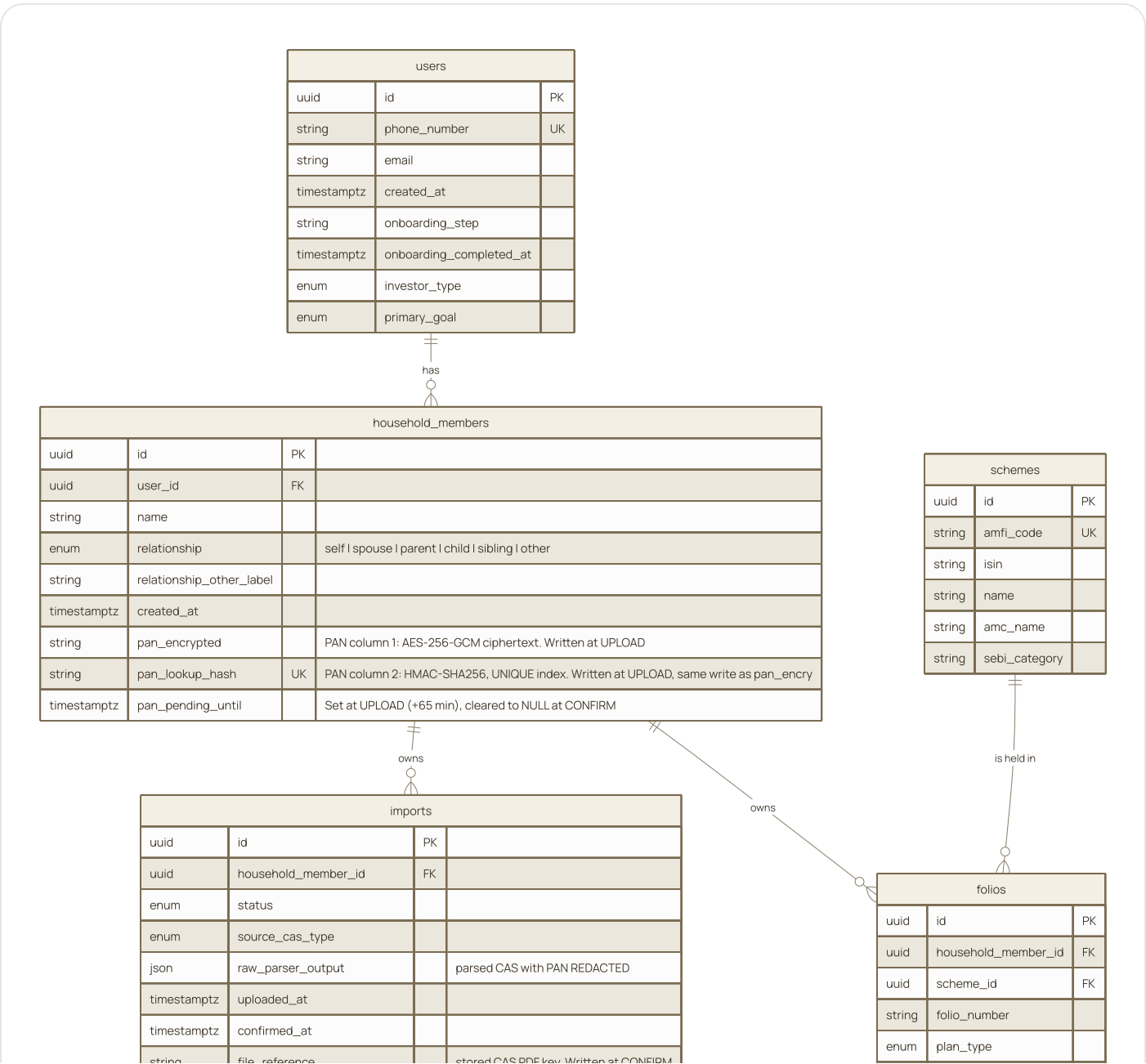
A member's PAN is always in one of three states: empty, pending or permanent.

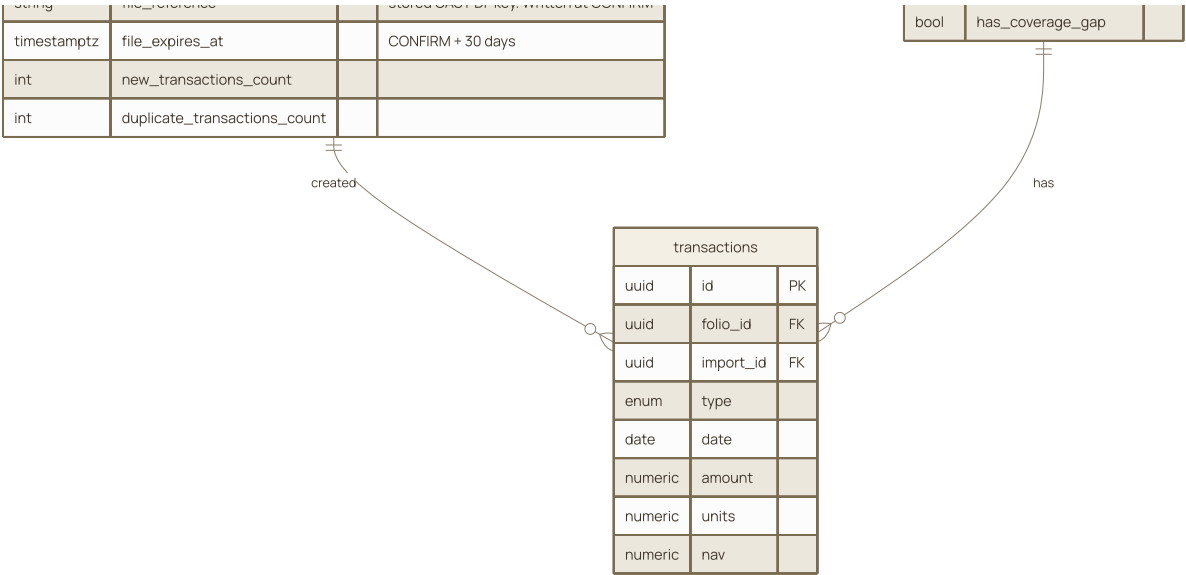


The emailed request-CAS path (/cas-imports) has no review screen, so it goes straight from **Empty** to **Permanent**. The current app screens don't use it.

ER diagram: the tables an import touches

The PAN lives on `household_members` . The unique index on `pan_lookup_hash` enforces one PAN, one member, across all of Unifolio, and a pending PAN holds that slot too.





Where the PAN is, and isn't, stored

PLACE	PAN STORED?	DETAIL
<code>household_members.pan_encrypted</code>	Yes	AES-256-GCM encrypted; readable only with the server key.
<code>household_members.pan_lookup_hash</code>	Yes, as a fingerprint	One-way HMAC-SHA256; cannot be turned back into the PAN.
<code>imports.raw_parser_output</code>	No	The PAN is removed before the parsed CAS is saved.
Stored CAS PDF (<code>imports.file_reference</code>)	Inside the PDF	As printed on the statement; deleted after 30 days.
Any API response or screen	No	Only the masked form, e.g. L*****R.
Server logs	No	Logs name the member id only.
Server memory (review session)	Yes, raw	For at most 60 minutes, until Confirm, Discard or expiry.

Source: Docs/orchestration/pan-storage-flow-and-er-diagram.md · reflects the PAN-at-upload change of 24 Sep 2026 (pan_claims.py, migration 0016).

